ABSTRACT

Cross-attention maps in text-to-image diffusion models are routinely treated as explanations for how text tokens influence image generation. We test this assumption directly. Across 16,302 controlled embedding-space interventions on 680 COCO prompts using Stable Diffusion XL [**?**], we find that attention maps are poor spatial predictors of where interventions take effect (SF-IoU-HR ≈ 0.17), and that the standard Attention-Delta Correlation (ADC) metric is partially tautological by construction. We introduce two decoupled metrics Predictive ADC (P-ADC) and Latent-Delta Correlation (L-ADC) that remove this circularity. The strongest finding is structural: *concrete-perceptual* attributes (color, material) concentrate target-concept attention roughly three times more than *abstract-stylistic* attributes (style, atmospheric effect), despite producing comparable image-level changes (LPIPS). This asymmetry holds across all intervention strengths and temporal windows, pointing to a mechanistic difference in how SDXL's cross-attention processes these two attribute classes.

1. INTRODUCTION

Cross-attention maps in diffusion models nominally show which text tokens influence each spatial position during denoising. This interpretation underpins widely-used methods: Prompt-to-Prompt editing [**?**], Attend-and-Excite [**?**], and DAAM [**?**] all either read model behavior from attention maps or directly modify them to achieve desired outputs. More recent work has proposed to use attention as a probe for implicit reasoning and semantic layout inside generation pipelines [**??**].

But faithfulness is not the same as correlation. A metric that compares the attention map computed *during* an intervention with the change *from* that same intervention is not an independent test both quantities share the same causal source. And a map that correlates with output change under ideal conditions may still fail to localize *where* the change happens spatially.

We construct a clean diagnostic. We apply training-free CLIP embedding-arithmetic interventions [**?**] to SDXL [**?**] and measure five properties of the resulting attention maps against observed image changes at scale, across six intervention strengths, four temporal windows, three prompt categories, and four attribute semantic types.

Summary of findings.

- Attention redistribution follows a dose–response curve with intervention strength (ACS, $r = -0.218$), saturating near $\alpha = 1.5$.
- The standard ADC metric is partially tautological; our decoupled P-ADC is statistically significant ($p < 10^{-90}$) but has near-zero effect size (mean ≈ 0.10, std ≈ 0.60).
- Attention maps overlap with changed image regions only $\approx$17% of the time (SF-IoU-HR).
- Concrete-perceptual attributes (Color, Material) produce attention concentration $\approx 3\times$ stronger than abstract-stylistic attributes (Style, Effect) at matched LPIPS.

2. RELATED WORK

Attention as explanation. **?** showed that attention weights are not reliable explanations in NLP; **?** partially contested this. For vision transformers, **?** proposed relevancy maps to improve on raw attention. In diffusion models, **?** (DAAM) used upsampled cross-attention as spatial explanations and found modest but non-trivial IoU with segmentation masks—a result our work replicates and contextualizes at much larger scale.

A crucial distinction, underscored by **?**, is that *cross-attention* and *self-attention* play fundamentally different roles in diffusion models: cross-attention handles semantic *injection* (binding text tokens to spatial features), while self-attention governs geometric layout and shape preservation. **?** exploit exactly this division—they transfer structure via self-attention features while allowing cross-attention to control appearance. This division of labor is a key reason why our SF-IoU-HR metric (≈ 0.17) is low: we are measuring cross-attention against a spatial outcome that is partly determined by a different attention stream.

Editing via attention manipulation. **?** showed that injecting cross-attention maps from a reference generation enables near-perfect structural consistency. **?** directly optimized attention to improve attribute binding. **?** and **?** used attention features for real-image inversion and structure transfer. **?** manipulated concept-token attention for instructable latent editing. **?** and **?** use concept-level attention routing for controlled concept editing.

Semantic latent structure. **?** found a semantic latent space (h-space) in diffusion models amenable to arithmetic manipulation. **?** analyzed how implicit assumptions encoded in the model can be re-written through attention-layer modifications. **?** probed what linguistic concepts the internal representations of diffusion models encode, finding that early diffusion layers are highly semantic.

Mechanistic interpretability in generative models. **?** established the mathematical framework for circuit analysis in transformers, and **?** proposed automated circuit discovery via activation patching in LLMs. These techniques are now being ported to diffusion models directly. **?** identify specific cross-attention heads in diffusion transformers responsible for spatial-relation generation via causal activation patching. **?** apply the same framework to 3D diffusion transformers, diagnosing and stabilizing trajectory failures. Our work is complementary: where these studies perform circuit-level causal patching, we provide a large-scale *observational* diagnostic that characterizes when and why cross-attention concentration is meaningful—a necessary precursor to knowing which circuits to patch.

3. Experimental Setup

Model. All experiments use `stabilityai/stable-diffusion-xl-base-1.0` (SDXL) [**?**], 50 DDIM [**?**] steps, guidance scale 7.5, seed 42, `float16`. Cross-attention probabilities are extracted via hooks at every denoising step (verified: $\sum_j p_{ij} = 1.0$, MAE $< 10^{-6}$).

Interventions. For each prompt we identify a *target concept* (a noun) and an *injection attribute*. The attribute is injected via CLIP embedding arithmetic:

$$\delta \;=\; \mathrm{Enc}(\text{attr}) - \mathrm{Enc}(\text{concept}),$$

scaled by strength $\alpha \in \{0.25, 0.50, 0.75, 1.0, 1.5, 2.0\}$ and applied only within a temporal window of denoising steps (four windows: 50–35, 35–20, 20–5, 45–5). No fine-tuning is performed.

Dataset. We sample 680 COCO 2017 captions [**?**] spanning three categories *Simple* (50 prompts), *Compositional* (451), and *Conflicting* (158). Twenty attributes are grouped into four semantic clusters: *Color* (blue, red, golden, silver, neon green), *Material* (glass, stone, wooden, metallic, velvet), *Style* (baroque, cyberpunk, futuristic, minimalist, ancient), *Effect* (fiery, frozen, glowing, misty, shadowy). Every prompt $\times$ attribute pair receives all 24 conditions (6 strengths $\times$ 4 windows), yielding **16,302 total interventions**.

Metrics.

- **ACS** (Attention Concentration Score): relative change in target-token attention mass, $\mathrm{ACS} = (\bar{a}^{\mathrm{interv}} - \bar{a}^{\mathrm{base}})/\bar{a}^{\mathrm{base}}$. Negative $\Longrightarrow$ attention *disperses* from the target.
- **ADC** (Attention-Delta Correlation): Pearson r between per-step intervention attention and per-step attention-delta. Retained for comparison; note both quantities derive from the same forward pass, introducing circularity.
- **P-ADC** (*ours*): Pearson r between *baseline-pass* attention (pre-intervention) and the per-step change induced by the intervention. Predictor and outcome are from separate forward passes.
- **L-ADC** (*ours*): Pearson r between baseline attention and per-step latent L2-divergence between baseline and intervention trajectories.
- **SF-IoU-HR** (DAAM-style [**?**]): IoU between the attention heatmap upsampled to image resolution and the pixel-change mask. Also computed at native resolution (SF-IoU).

4. Results

4.1. Dose-Response and the Limits of ACS

Table 1 summarizes metrics across intervention strengths. ACS scales monotonically with α ($r = -0.218$, $p < 10^{-3}$, Cohen's $d = 1.12$ for $\alpha = 0.25$ vs. $\alpha = 2.0$), then saturates above $\alpha = 1.5$. LPIPS increases monotonically.

What does not scale is predictive power. P-ADC stays near zero (mean ≈ 0.097, std ≈ 0.60) across all strengths. It is statistically nonzero ($t = 18.9$, $p < 10^{-90}$, $N = 15{,}798$) but the large standard deviation makes it an unreliable per-sample predictor. SF-IoU-HR plateaus at 0.165–0.180: attention heatmaps overlap with changed pixels only $\approx$17% of the time.

Table 1: Metrics by intervention strength α. Means over all prompts, attributes, and windows.

α	ACS	P-ADC	L-ADC	SF-IoU-HR	LPIPS	N
0.25	-0.105	0.100	-0.099	0.165	0.067	2636
0.50	-0.174	0.098	-0.084	0.166	0.096	2634
0.75	-0.217	0.097	-0.072	0.168	0.120	2632
1.00	-0.243	0.098	-0.063	0.170	0.142	2632
1.50	-0.259	0.099	-0.047	0.176	0.182	2632
2.00	-0.248	0.096	-0.035	0.180	0.220	2632

Table 2: Metrics by attribute cluster. Color and Material concentrate target attention $\sim 3\times$ more than Style and Effect, at matched LPIPS.

Cluster	ACS	P-ADC	SF-IoU-HR	LPIPS
Color	-0.306 ± 0.13	0.090 ± 0.60	0.177	0.150
Material	-0.349 ± 0.16	0.096 ± 0.59	0.167	0.132
Style	-0.106 ± 0.18	0.088 ± 0.58	0.179	0.138
Effect	-0.098 ± 0.19	0.118 ± 0.62	0.163	0.131

4.2. Attribute Type Predicts Attention Engagement

The clearest signal is not strength- or timing-dependent. As shown in Table 2 and Figure 1b, *attribute semantic type* is the dominant predictor of ACS. Color and Material attributes consistently show strong attention concentration (ACS ≈ -0.31 and -0.35, respectively). Style and Effect attributes produce roughly **three times weaker** attention shifts (ACS ≈ -0.11 and -0.10), despite similar LPIPS values (0.130–0.150 vs. 0.131–0.138).

This asymmetry replicates across all prompt categories, strength levels, and temporal windows. It is not driven by differences in LPIPS: the image-level perturbation is comparable, but only concrete-perceptual attributes trigger proportional attention concentration. One account: concrete token concepts ("blue", "wooden") correspond to tighter, more localized CLIP embeddings, so the steering direction δ lands in a region of embedding space that maps to focused cross-attention redistribution. Abstract concepts ("baroque", "misty") have diffuse embeddings; the resulting δ changes the latent trajectory without producing a correspondingly localized attention shift. This is consistent with **?**, who show that excessive cross-attention *spread and overlap* directly causes attribute binding failures and entity-missing artefacts—our Style and Effect attributes, with their near-zero ACS, sit precisely in this failure-prone regime. We leave direct testing of this account via text-encoder probing or targeted attention knockout as future work.

Table 3: Metrics by temporal intervention window. DDIM step ranges out of 50 total steps.

Window	ACS	P-ADC	SF-IoU-HR	LPIPS
Early (50–35)	-0.223	-0.065	0.156	0.242
Mid (35–20)	-0.203	$+0.275$	0.170	0.083
Late (20–5)	-0.199	-0.076	0.177	0.042
Full (45–5)	-0.206	$+0.258$	0.177	0.183

4.3. Temporal Window Effect

Table 3 shows metrics by intervention window. ACS is stable (range 0.199–0.223) across all four windows, suggesting that *when* the intervention is applied has little effect on how much attention concentrates. SF-IoU-HR shows a small but consistent gradient: 0.156 in the early window (steps 50–35, structural phase) rising to 0.177 in the late and full windows. LPIPS varies strongly: early interventions reshape global structure (LPIPS $= 0.242$) while late interventions affect fine detail only (LPIPS $= 0.042$).

5. Discussion

The ADC metric has a structural problem. ADC values near -1.0 for Color and Material clusters arise because the metric correlates attention *from* the intervention pass with attention deltas *from* the same pass. P-ADC breaks this by using baseline-pass attention as the predictor. The resulting near-zero P-ADC (mean ≈ 0.10) is itself a result: knowing where the model naturally attends to a concept tells us little about where an intervention will concentrate its effect per-sample.

Attention maps are spatially imprecise. SF-IoU-HR ≈ 0.17 means roughly 83% of high-attention pixels do not coincide with the pixels that actually changed. This is consistent with DAAM's findings on SD 1.4 [**?**] and with **?**'s observation that internal diffusion representations are only loosely spatially grounded. A structural explanation follows from **?**: cross-attention injects *semantic* signal into the residual stream, but the actual spatial realization of that signal is mediated downstream by self-attention layers. Measuring cross-attention against a pixel-change mask conflates two separate computations. Interventions *work*—LPIPS increases monotonically—but the cross-attention map is a poor spatial readout of where the effect lands precisely because spatial structure is handled elsewhere.

The attribute-type asymmetry is the most actionable finding. Color and Material: $|\text{ACS}| \approx 0.33$. Style and Effect: $|\text{ACS}| \approx 0.10$. LPIPS $\approx$ 0.13–0.15 for both groups. This matches the prediction of **?** that concrete visual concepts occupy more compact latent neighborhoods. A direct mechanism test via targeted attention knockout for Color vs. Style tokens, or CLIP embedding-space geometry analysis, is a natural next step. The causal patching methodology for such experiments in diffusion models has been demonstrated by **?** and **?**, providing a ready-made experimental framework.

Limitations. (1) Single model (SDXL base 1.0); cross-model generalization to SD 3 [**?**] or FLUX is untested. (2) P-ADC establishes predictive correlation, not causality. (3) COCO prompts may not represent real-world generation requests.

6. Conclusion

We ran 16,302 controlled interventions on SDXL and measured whether cross-attention maps faithfully predict intervention effects. They do not, spatially (SF-IoU-HR ≈ 0.17), and the standard ADC metric overstates their predictive power via circularity. The most replicable finding is structural: Color and Material attributes concentrate target-concept attention three times more strongly than Style and Effect attributes, at matched image-level perturbation. This asymmetry is a concrete, testable hypothesis about how SDXL's cross-attention differentially processes concrete vs. abstract token meanings, and motivates targeted causal ablation experiments. Code and data: `https://anonymous.4open.science/r/diffusion-detective`.

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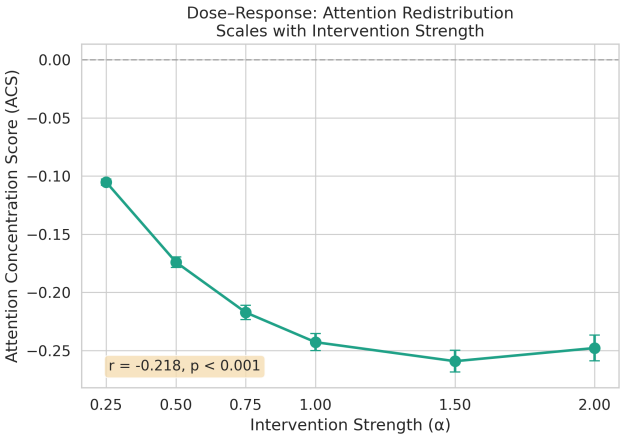
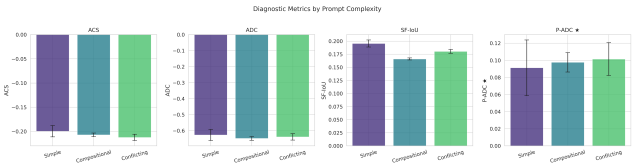


(a) ACS vs. α. Error bars = 95% CI. Monotonic; saturates at $\alpha \approx 1.5$ ($r = -0.218$, Cohen's $d = 1.12$).



(b) ACS by attribute cluster and prompt category. Color/Material gap ($\sim 3\times$) dominates any category-level effect.

Figure 1: **Key findings. (a)** Attention redistribution scales with strength and saturates a dose–response curve. **(b)** Attribute semantic type, not prompt category, is the primary predictor of attention engagement.

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A. Full Metric Tables

Table 4: All metrics by intervention strength (mean; std in parentheses).

α	ACS	ADC	P-ADC	L-ADC	SF-IoU	SF-IoU-HR	LPIPS
0.25	-0.105	-0.610	0.100	-0.099	0.162	0.165	0.067
	(0.070)	(0.517)	(0.594)	(0.793)	(0.110)	(0.096)	(0.072)
0.50	-0.174	-0.690	0.098	-0.084	0.165	0.166	0.096
	(0.121)	(0.528)	(0.596)	(0.799)	(0.110)	(0.098)	(0.089)
0.75	-0.217	-0.695	0.097	-0.072	0.169	0.168	0.120
	(0.160)	(0.566)	(0.597)	(0.801)	(0.110)	(0.099)	(0.102)
1.00	-0.243	-0.677	0.098	-0.063	0.172	0.170	0.142
	(0.192)	(0.611)	(0.599)	(0.802)	(0.109)	(0.098)	(0.114)
1.50	-0.259	-0.625	0.099	-0.047	0.179	0.176	0.182
	(0.245)	(0.685)	(0.598)	(0.801)	(0.107)	(0.097)	(0.130)
2.00	-0.248	-0.569	0.096	-0.035	0.183	0.180	0.220
	(0.291)	(0.741)	(0.600)	(0.801)	(0.104)	(0.095)	(0.141)

Table 5: All metrics by prompt category.

Category	ACS	P-ADC	L-ADC	SF-IoU-HR	LPIPS	N_{prompts}
Compositional	-0.207	0.098	-0.046	0.165	0.137	451
Conflicting	-0.212	0.102	-0.022	0.178	0.138	158
Simple	-0.200	0.091	-0.394	0.187	0.141	50

Table 6: Full metrics by temporal window.

Window	ACS	ADC	P-ADC	L-ADC	SF-IoU-HR	LPIPS
Early (50–35)	-0.223	-0.701	-0.065	$+0.033$	0.156	0.242
Mid (35–20)	-0.203	-0.626	$+0.275$	-0.093	0.170	0.083
Late (20–5)	-0.199	-0.674	-0.076	-0.202	0.177	0.042
Full (45–5)	-0.206	-0.577	$+0.258$	-0.005	0.177	0.183

B. Statistical Tests

- ACS $\neq 0$: significant at all strengths ($p < 10^{-100}$).
- P-ADC $\neq 0$: $t = 18.9,\ p < 10^{-90},\ N = 15{,}798$. Mean $= 0.098$, std $= 0.60$; effect size is small.
- $r(\text{ACS}, \alpha) = -0.218,\ p < 10^{-3}$ (dose–response).
- $r(|\text{ACS}|, \text{LPIPS}) = +0.63,\ p < 10^{-100}$.
- Cohen's d (ACS: $\alpha{=}0.25$ vs. $\alpha{=}2.0$) $= 1.12$.
- ADC $\neq 0$: $p < 10^{-100}$; partially tautological (see Section 3).

C. Implementation Details

Hardware. Single NVIDIA V100 (16 GB). Throughput ≈ 45 interventions per GPU-hour.
SDXL settings. `stabilityai/stable-diffusion-xl-base-1.0`, 50 DDIM steps, guidance scale 7.5, seed 42, `float16`.
Attention extraction. Hooks on all `CrossAttention` modules; probability rows verified to sum to 1.0 within MAE $< 10^{-6}$.

CLIP metric. `openai/clip-vit-base-patch32` [**?**]. Δ-CLIP uses the directional formulation from StyleGAN-NADA [**?**].
LPIPS. VGG backbone, `torchmetrics` implementation [**?**].
SF-IoU-HR. Attention maps mean-pooled across heads and steps, bilinearly upsampled to 1024×1024 (DAAM protocol [**?**]), binarized at 75th percentile. Same percentile threshold applied to the pixel-change mask.